

Iterated Dyadic Schur Continuation of a Physical Spectral Count

A Hierarchical Resolvent Certificate from 2048 to 8192 Cells

at a Quadratic Band-Merging Map

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July 2026

Abstract

A preceding computer-assisted theorem proved that exact stored Perron/parity-extracted physical two-step matrices on 2048 and 4096 midpoint cells each have one eigenvalue inside a prescribed complex circle. Its direct coarse resolvent atlas did not by itself provide an efficient route to the next grid. We give a second rigorous dyadic continuation, at fixed Gaussian width $\sigma = 10^{-2}$, and introduce a hierarchical resolvent estimate that avoids a direct 4096-dimensional inverse atlas.

For the $4096 \rightarrow 8192$ split, componentwise outward residuals around four rank-96 centers certify the coarse-consistency, coarse-to-detail, detail-to-coarse, and detail blocks by

$$4.97036 \times 10^{-5}, \quad 5.66125 \times 10^{-3}, \quad 6.86176 \times 10^{-3}, \quad 3.26337 \times 10^{-5}.$$

The resulting second-level effective perturbation is at most 1.52962×10^{-4} , so an A_{4096} resolvent bound below 6537.58 is sufficient. We tighten the inherited A_{2048} atlas from 170 to 183 rigorous centers and form an exact 324-leaf rational contour partition. An exact Schur block-inverse formula propagates each leafwise A_{2048} bound through the first refinement. The worst propagated A_{4096} resolvent is 4843.82, and the final matrix Rouché product is $0.740919 < 1$. Hence, for the exact archived binary64 matrices,

$$N_{\Gamma}(A_{2048}) = N_{\Gamma}(A_{4096}) = N_{\Gamma}(A_{8192}) = 1.$$

This is a finite stored-matrix theorem at one noise scale. It does not prove a continuum limit, an all-dimensions induction, a zero-noise limit, a zeta-zero identification, a Hilbert–Pólya construction, or the Riemann hypothesis.

1 Introduction

The previous nested-grid certificate [7] answered one discretization-stability question: a contour count proved at dimension 2048 survives the refinement to 4096. Repeating that proof naively at the next scale would require a direct atlas of rigorous inverses for a 4096-dimensional nonnormal matrix. Such an atlas is possible in principle, but it discards the exact coarse/detail structure already certified at the first refinement.

This paper asks a sharper question. Can the first Schur decomposition be used not only to transfer a count, but also to transfer a *resolvent bound* strong enough to close the next count comparison? The answer is yes for the exact stored matrices studied here. The proof chain is

$$\begin{aligned} \text{direct } A_{2048} \text{ resolvent centers} &\longrightarrow \text{first Schur effective inverse} \longrightarrow \text{full } A_{4096} \text{ resolvent} \\ &\longrightarrow \text{second Schur effective comparison} \longrightarrow N_{\Gamma}(A_{8192}) = N_{\Gamma}(A_{4096}). \end{aligned}$$

The distinction matters because these transfer matrices are strongly nonnormal. Eigenvalue displacements alone do not control contour counts; resolvent amplification is the relevant quantity [4, 6]. The new estimate keeps that amplification visible at both refinement levels.

The principal contributions are as follows.

1. We certify the exact $4096 \rightarrow 8192$ coarse/detail blocks by componentwise factor-graph residuals. Relative to the first split, the two consistency/detail blocks are approximately one quarter as large and the cross channels approximately one half as large.
2. We prove a norm-preserving block-inverse propagation formula. It turns a leafwise A_{2048} resolvent upper M_0 into a rigorous A_{4096} upper without constructing a 4096-dimensional inverse.
3. We tighten the inherited direct atlas with only 13 additional centers. The resulting exact rational partition closes both the first effective inverse gate and the second matrix Rouché gate.
4. We bitwise identify the A_{4096} factor object used here with the one certified in the preceding theorem, thereby obtaining the three-scale count chain.

The observed scaling is encouraging, but this paper does not promote two finite ratios to an asymptotic law. An induction would require uniform block estimates and a uniformly controlled hierarchical resolvent atlas, neither of which is proved here.

2 Stored matrices, contour, and inherited count

Let P_m be the archived row-normalized folded Gaussian midpoint matrix on m positive cells. Let

$$X_m, Y_m \in \mathbb{R}^{m \times 2}, \quad \Lambda_m \in \mathbb{R}^{2 \times 2}$$

be the stored right modes, left modes, and peripheral eigenvalue diagonal. The Perron/parity-extracted one-step and physical two-step matrices are

$$U_m = P_m - X_m \Lambda_m Y_m^T, \quad A_m = U_m^2. \quad (1)$$

Every binary64 entry in these factors is treated as an exact real number. We write

$$A_0 = A_{2048}, \quad A_1 = A_{4096}, \quad A_2 = A_{8192}.$$

The positively oriented counting circle is

$$\Gamma : \quad z = z_0 + r e^{i\theta}, \quad 0 \leq \theta \leq 2\pi, \quad (2)$$

with stored values

$$z_0 = -0.3233504401504541 - 0.5508412474453575i, \quad r = 0.2624987592858511. \quad (3)$$

The distance from the origin to the closed counting disk satisfies

$$\rho_0 = |z_0| - r \geq 0.376235604149098. \quad (4)$$

For a matrix A , let $N_\Gamma(A)$ denote the algebraic number of eigenvalues in the interior of Γ .

Proposition 2.1 (Inherited stored count). *For the exact archived factors at $\sigma = 10^{-2}$,*

$$N_\Gamma(A_0) = N_\Gamma(A_1) = 1.$$

The A_1 sparse matrix, right and left peripheral modes, and peripheral values consumed in the present snapshot are bitwise identical to those in the preceding certificate.

Proof. The count is the main theorem of Wang [7], built on the exact packet-pair count of Wang [9]. The present replay hashes the sparse data, indices, row pointers, right modes, left modes, and peripheral values independently. All six hashes agree with the inherited A_1 object, and the inherited snapshot and theorem hashes also agree with the archive. $\square$

We distinguish exact algebra, rigorous stored-binary64 certificates, and floating diagnostics. Only the first two enter the theorem. Sparse eigensolves are reported later solely to localize the counted eigenvalue.

3 Exact dyadic coordinates at two levels

For a dyadic refinement from n to $2n$, define replication and alternating-detail injections by

$$(Jx)_{2k} = (Jx)_{2k+1} = x_k, \quad (Ky)_{2k} = y_k, \quad (Ky)_{2k+1} = -y_k.$$

Define the corresponding restrictions

$$R = \frac{1}{2}J^T, \quad S = \frac{1}{2}K^T,$$

and set

$$T = [J \ K], \quad T^{-1} = \begin{bmatrix} R \\ S \end{bmatrix}.$$

These identities are exact over the stored reals. Moreover $T = \sqrt{2}Q$ for an orthogonal Q , so

$$T^{-1}MT = Q^T M Q, \quad \|T^{-1}MT\|_2 = \|M\|_2. \quad (5)$$

Thus the unnormalized integer-valued coordinate maps introduce no condition number loss in the Euclidean norm.

At refinement level $j = 1, 2$, the exact coordinate block form is

$$T_j^{-1}A_jT_j = \begin{pmatrix} A_{j-1} + E_j & B_j \\ C_j & D_j \end{pmatrix}. \quad (6)$$

Here C_j maps coarse coordinates to detail coordinates, while B_j maps detail coordinates back to coarse coordinates. Whenever $z - D_j$ is invertible, define

$$R_{D,j}(z) = (z - D_j)^{-1}, \quad (7)$$

$$\Delta_j(z) = E_j + B_j R_{D,j}(z) C_j, \quad (8)$$

$$F_j(z) = z - A_{j-1} - \Delta_j(z). \quad (9)$$

The exact Schur determinant identity is

$$\det(z - A_j) = \det(z - D_j) \det F_j(z). \quad (10)$$

If $\|D_j\|_2 < \rho_0$, every eigenvalue of D_j lies in the origin-centered disk of radius $\|D_j\|_2$, which is disjoint from the counting disk. In that case

$$\sup_{z \in \Gamma} \|R_{D,j}(z)\|_2 \leq d_j := \frac{1}{\rho_0 - \|D_j\|_2}, \quad (11)$$

and

$$\sup_{z \in \Gamma} \|\Delta_j(z)\|_2 \leq \varepsilon_j := \|E_j\|_2 + \|B_j\|_2 d_j \|C_j\|_2. \quad (12)$$

4 Certified second-level block bounds

The proof objects for each block are stored rank-96 factors $L\Sigma V^T$. They are not assumed exact singular decompositions. The following elementary estimate explains how they enter the certificate.

Lemma 4.1 (Low-rank center plus full residual). *Let H be an exact stored block, and let L, Σ, V be stored factors. If*

$$\|L^T L - I\|_F \leq \delta_L, \quad \|V^T V - I\|_F \leq \delta_V,$$

then

$$\|H\|_2 \leq \sqrt{1 + \delta_L} \|\Sigma\|_2 \sqrt{1 + \delta_V} + \|H - L\Sigma V^T\|_F. \quad (13)$$

Proof. The Gram estimates imply $\|L\|_2 \leq \sqrt{1 + \delta_L}$ and $\|V\|_2 \leq \sqrt{1 + \delta_V}$. Apply the triangle inequality, submultiplicativity, and $\|R\|_2 \leq \|R\|_F$. $\square$

Every matrix action used to form the residual in (13) is evaluated by componentwise outward arithmetic. The reported Frobenius residual contains both the residual center and its complete radius. Standard verified-floating-point principles are described in Higham [3], Rump [5]; the factor-graph implementation is inherited from Wang [8].

Table 1: Certified block norms at the two dyadic levels. The last column is the full second-level Frobenius residual after subtracting the stored rank-96 center.

block	first-level upper	second-level upper	ratio	second residual
E	$1.98877177 \times 10^{-4}$	$4.97035044 \times 10^{-5}$	0.249921	7.4855×10^{-10}
C	$1.13214379 \times 10^{-2}$	$5.66124305 \times 10^{-3}$	0.500046	1.5175×10^{-8}
B	$1.37226417 \times 10^{-2}$	$6.86175172 \times 10^{-3}$	0.500031	7.1604×10^{-9}
D	$1.30522077 \times 10^{-4}$	$3.26336986 \times 10^{-5}$	0.250024	3.3016×10^{-10}

For the second split, (11) gives

$$d_2 \leq 2.658139564401696. \quad (14)$$

The second Schur self-energy and total effective perturbation satisfy

$$\|B_2\|_2 d_2 \|C_2\|_2 \leq 1.0325820704184003 \times 10^{-4}, \quad (15)$$

$$\varepsilon_2 \leq 1.5296171140743076 \times 10^{-4}. \quad (16)$$

Consequently, a contour-wide A_1 resolvent bound below

$$\varepsilon_2^{-1} \geq 6537.583757391333 \quad (17)$$

is sufficient for the second matrix Rouché comparison.

5 Hierarchical propagation of the 4096-dimensional resolvent

The threshold in (17) could be tested by direct 4096-dimensional inverse certificates. Instead we propagate the existing direct A_0 atlas through the first exact block decomposition.

Proposition 5.1 (Schur resolvent propagation). *Fix $z \in \Gamma$. Suppose*

$$\|(z - A_0)^{-1}\|_2 \leq M_0, \quad M_0 \varepsilon_1 < 1.$$

Let d_1 be the first detail-resolvent upper from (11). Then

$$\begin{aligned} \|(z - A_1)^{-1}\|_2 &\leq M_1(M_0) := d_1 \\ &\quad + \sqrt{1 + (d_1 \|C_1\|_2)^2} \sqrt{1 + (d_1 \|B_1\|_2)^2} \frac{M_0}{1 - M_0 \varepsilon_1}. \end{aligned} \quad (18)$$

Proof. Write $R_0 = (z - A_0)^{-1}$. By (8)–(9),

$$F_1 = (z - A_0)(I - R_0 \Delta_1(z)).$$

The Neumann bound gives

$$\|F_1^{-1}\|_2 \leq \frac{M_0}{1 - M_0 \varepsilon_1}. \quad (19)$$

The exact inverse of the first block matrix factors as

$$\begin{aligned} &\left(z - \begin{pmatrix} A_0 + E_1 & B_1 \\ C_1 & D_1 \end{pmatrix} \right)^{-1} \\ &= \begin{bmatrix} I \\ R_{D,1} C_1 \end{bmatrix} F_1^{-1} \begin{bmatrix} I & B_1 R_{D,1} \end{bmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & R_{D,1} \end{pmatrix}. \end{aligned} \quad (20)$$

The column and row graph factors obey

$$\begin{aligned} \left\| \begin{bmatrix} I \\ R_{D,1} C_1 \end{bmatrix} \right\|_2 &\leq \sqrt{1 + (d_1 \|C_1\|_2)^2}, \\ \left\| \begin{bmatrix} I & B_1 R_{D,1} \end{bmatrix} \right\|_2 &\leq \sqrt{1 + (d_1 \|B_1\|_2)^2}. \end{aligned}$$

Combine these estimates with (19). Finally, (5) identifies the coordinate-space norm with the ambient A_1 resolvent norm. $\square$

The first-level certificate supplies

$$d_1 \leq 2.658831394912469, \quad \varepsilon_1 \leq 6.119533154024968 \times 10^{-4}. \quad (21)$$

The two graph factors in (18) are bounded by

$$1.000452956430038, \quad 1.000665399669618.$$

Thus the only remaining task is a sufficiently tight leafwise bound for the direct A_0 resolvent.

6 A tightened direct atlas and two simultaneous gates

At a rigorous direct center ζ , let

$$\left\|(\zeta - A_0)^{-1}\right\|_2 \leq M_\zeta.$$

For a circular-arc enclosure contained in a disk satisfying $M_\zeta \delta < 1$, the resolvent identity yields

$$\sup_{|z-\zeta|\leq\delta} \left\|(z - A_0)^{-1}\right\|_2 \leq \frac{M_\zeta}{1 - M_\zeta \delta}. \quad (22)$$

The inherited archive contains 170 such centers. We request the stronger first-level target

$$M_0 \varepsilon_1 < 0.75, \quad M_0 < \frac{0.75}{\varepsilon_1} = 1225.583686080214 \dots \quad (23)$$

Only 13 additional direct centers are needed. Adaptive exact bisection then produces 324 rational leaves, with maximum refinement level eight and no unresolved component. The leaves form an exact partition of one contour turn.

Table 2: Outward-rounded maxima over the exact 324-leaf propagated atlas.

quantity	certified upper
transported A_0 resolvent M_0	1221.3793571000042
first effective product $M_0 \varepsilon_1$	0.7474271469415178
first effective inverse $\left\ F_1^{-1}\right\ _2$	4835.750724236383
propagated A_1 resolvent M_1	4843.819104431207
second continuation product $M_1 \varepsilon_2$	0.7409188599618061

Both products are evaluated leaf by leaf; the maxima need not occur on the same leaf. In particular, the final value in [table 2](#) lies well below one and the propagated A_1 bound lies below the independent threshold in [\(17\)](#).

7 The certified three-scale count

We use the finite-dimensional analytic matrix form of Rouché’s theorem [\[1, 2\]](#). If $A(z)$ is invertible on Γ , $B(z)$ is analytic on and inside the contour, and

$$\sup_{z \in \Gamma} \left\|A(z)^{-1}B(z)\right\|_2 < 1,$$

then A and $A + B$ have the same determinant zero count inside the contour.

Theorem 7.1 (Certified iterated dyadic physical count). *At $\sigma = 10^{-2}$, for the exact stored binary64 physical two-step matrices defined in [\(1\)](#),*

$$\boxed{N_\Gamma(A_{2048}) = N_\Gamma(A_{4096}) = N_\Gamma(A_{8192}) = 1.}$$

Proof. By [proposition 2.1](#), $N_\Gamma(A_1) = 1$. The second detail block satisfies

$$\|D_2\|_2 \leq 3.2633698567 \times 10^{-5} < \rho_0,$$

so $z - D_2$ is invertible on and inside the counting disk and $N_\Gamma(D_2) = 0$. Hence $\Delta_2(z)$ is analytic there.

On every leaf, [proposition 5.1](#) and the direct center transport (22) give a rigorous bound for $(z - A_1)^{-1}$. The exact rational partition covers all of Γ , and [table 2](#) gives

$$\sup_{z \in \Gamma} \|(z - A_1)^{-1} \Delta_2(z)\|_2 \leq 0.7409188599618061 < 1.$$

Matrix Rouché therefore shows that

$$N_\Gamma(F_2) = N_\Gamma(z - A_1) = 1.$$

The exact determinant factorization (10) yields

$$N_\Gamma(A_2) = N_\Gamma(D_2) + N_\Gamma(F_2) = 0 + 1 = 1.$$

Combining this with the inherited count proves the stated chain. $\square$

8 Finite-scale diagnostics and what they do not prove

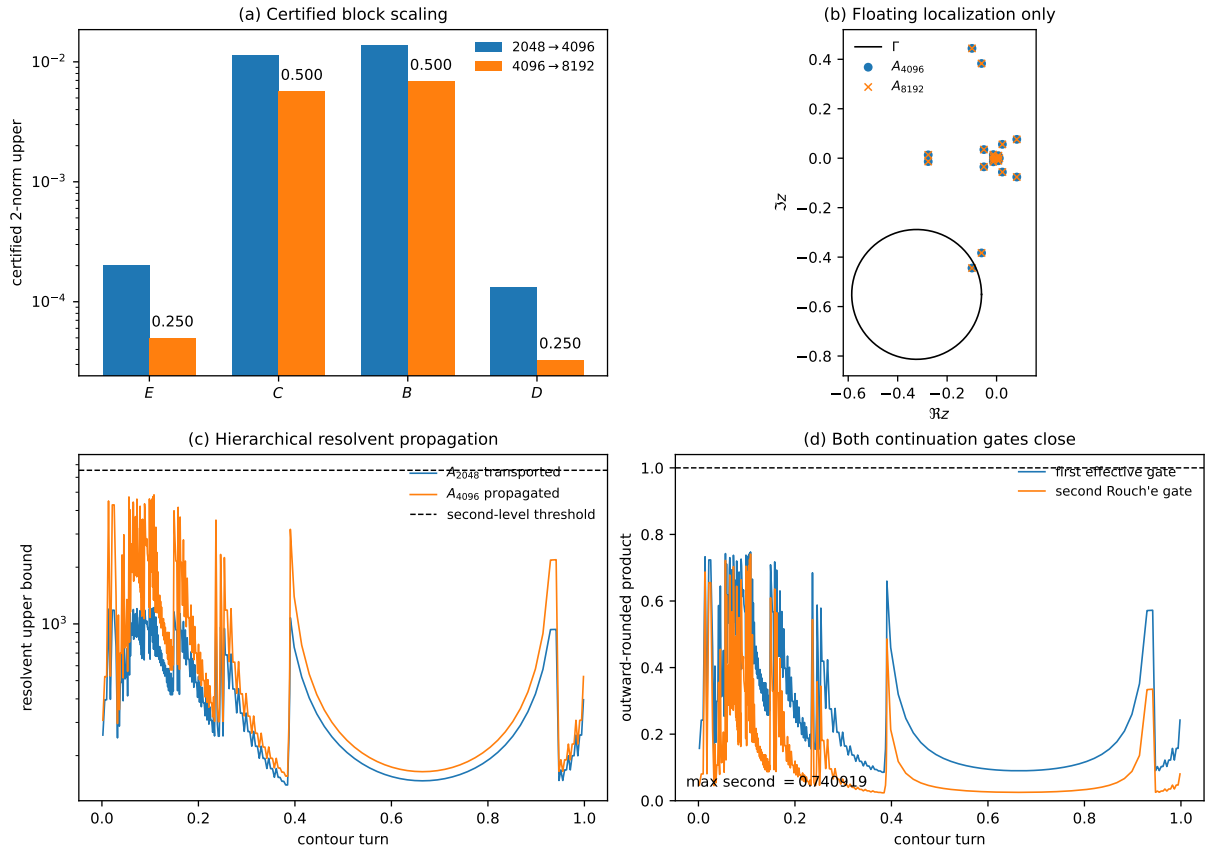


Figure 1: Second dyadic continuation diagnostics and certificates. (a) The second certified consistency and detail blocks are approximately one quarter of the first-level bounds, while the cross channels are approximately one half. (b) Floating sparse eigenvalues are shown only for localization; the count theorem does not use them. (c) The direct A_{2048} leaf bounds are propagated to A_{4096} by (18), remaining below the second-level threshold. (d) Both outward-rounded continuation products stay below one.

The floating pilot resolves one eigenvalue inside Γ at each of the two larger dimensions. Their approximate locations are

$$-0.0994035212675 - 0.4441920040123i$$

and

$$-0.0994142875185 - 0.4441895651018i,$$

with displacement 1.10391×10^{-5} . Residuals are below 1.3×10^{-15} . These values help visualize the theorem but do not certify the count.

The second-to-first block ratios are

$$0.249921, \quad 0.500046, \quad 0.500031, \quad 0.250024.$$

The self-energy and full effective perturbation ratios are respectively 0.249974 and 0.249957. This is consistent with a first-order decay of cross channels and a second-order decay of consistency/detail errors under one additional dyadic refinement. It is evidence for a possible multilevel estimate, not a proof of one. A genuine induction would have to control:

1. the block ratios uniformly over all sufficiently fine grids;
2. the growth of the recursively propagated nonnormal resolvent;
3. the number and quality of direct base-grid centers needed to cover the full contour; and
4. the relation of the stored finite matrices to a continuum operator.

The present theorem closes one additional finite level and isolates these remaining obstructions quantitatively.

9 Archive and reproducibility

The archive separates proof inputs from diagnostics.

- The exact A_{8192} sparse factor object is stored separately from four rank-96 center archives. Every file is below the repository's single-file limit. The hash of the original monolithic construction snapshot and every constituent array hash are retained.
- The second block certificate records factor Gram defects, singular scale uppers, full residual Frobenius uppers, and residual hashes.
- The direct center archive contains 170 inherited and 13 additional rigorous A_{2048} center bounds.
- The 324-leaf CSV ledger records exact rational endpoints, center transport data, first effective products, propagated A_{4096} bounds, and second continuation products.
- A final JSON certificate composes object identity, detail exclusion, inherited count, exact partition, and both Rouché gates.

The fast verification sequence is

```
python -m pytest -q -p no:cacheprovider
python experiments/build_final_certificate.py
python experiments/build_archive.py
MPLBACKEND=Agg python experiments/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
cp main.pdf iterated-dyadic-physical-count.pdf
python experiments/verify_archive.py
```


Recomputing the componentwise second-level residual certificate is the most expensive replay step. It is run by

```
OPENBLAS_NUM_THREADS=32 OMP_NUM_THREADS=32 \
python experiments/run_second_dyadic_block_certificate.py \
--chunk-size 128
```

The theorem is intentionally narrow. It compares exact finite stored binary64 matrices at one Gaussian width. It does not enclose discretization error relative to a continuum transfer operator, prove an arbitrary-grid limit, move σ toward zero, identify a zeta zero, construct a self-adjoint Hilbert–Pólya operator, or imply the Riemann hypothesis.

10 Conclusion

The $4096 \rightarrow 8192$ physical count continuation closes without a direct 4096-dimensional inverse atlas. The decisive step is the exact first-level block inverse, which turns a tightened direct A_{2048} atlas into a rigorous A_{4096} resolvent atlas. Combined with the quarter-scale second effective perturbation, this yields a final product $0.740919 < 1$ and the certified three-grid count chain

$$N_{\Gamma}(A_{2048}) = N_{\Gamma}(A_{4096}) = N_{\Gamma}(A_{8192}) = 1.$$

The result does not yet establish multilevel convergence, but it converts the next question from “can one afford another direct atlas?” to the more structural problem of proving uniform block decay and hierarchical resolvent control.

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